Volcano levels of End(E): prime-power points, isogeny walks, factoring

curve_traits, branch `claude/curve-traits-power-points`, 2026-10-22. Source of every number: `docs/curves/traits.json` on `main` (before) and on this branch (after), tabulated by `tabulate.py` into `levels.json`.

What ran

`curve_traits build` over the 2572 registry curves and 176 source curves, with the methods this branch adds:

- **Prime-power point test** (`points.rs`, `scalar_on_power_by_points`): Frobenius is the scalar λ on $E[\ell^j]$ only if $E[\ell^j] \subseteq E(\mathrm{GF}(q^m))$, $m = \mathrm{ord}(\lambda \bmod \ell^j)$; the test refutes or confirms each candidate λ with points of that group and Frobenius applied to their coordinates.
- **Quadratic twist for even k** in the depth-1 point test, after the untwisted test, so no earlier decision changes.
- **Isogeny walks** (`walk.rs`): cycles of horizontal l-isogenies, $l \le 19$, on j-invariants through Φ_l. A closed cycle of length r is a cyclic endomorphism of degree l^r whose conductor (Cornacchia) the conductor of End(E) divides, and r is exactly the order of a prime above l in the class group of End(E) (computed by composing quadratic forms); a walk that does not close bounds that order from below. The orders between $\mathbb{Z}[\pi]$ and $\mathcal{O}_K$ that survive decide the levels.
- **Class number 1**: $j(E) = j(\mathcal{O}_K)$ puts $\mathcal{O}_K$ in End(E).
- **Factor hints** for the discriminant cofactors: PARI, GMP-ECM and CADO-NFS (C99, C109), every factor proved prime by PARI's `isprime`.

Results (counts from the two files)

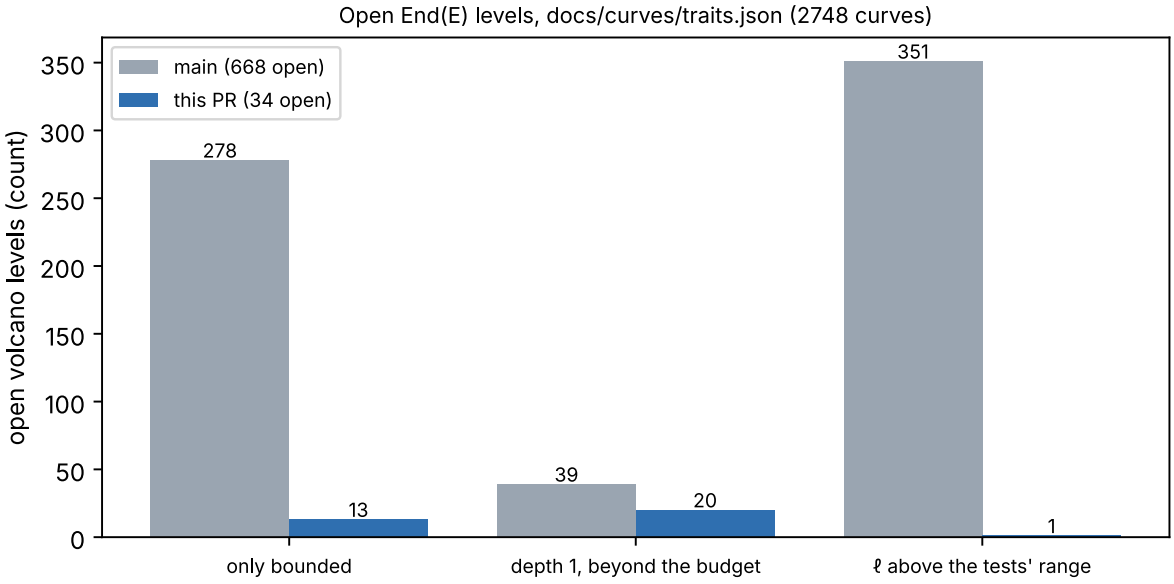


Figure 1: Open volcano levels by kind, `main` against this branch (all 2748 curves; counts, no sampling).

	main	this branch
End(E) established	2093	2700
End(E) only bounded	620	34
Δ not factored	35	14
open levels	668	34

No level decided on `main` changes value or loses its decision on this branch (checked level by level). `curve_traits check` reproduces the file. Build cost on the 4-core container: `curve_traits build` 22.0 CPU-minutes, 8.7 minutes wall (`main`, before its registry grew to 2572 curves, measured 18.0 CPU-minutes; not re-measured since).

What stays open, and why

All but two of the 34 open levels are on `GF(p^k)` curves whose `|d_K|` has 167 to 245 bits, so no walk can close: 19 at depth 1 for ℓ from 29 to 479, where `k = ord(λ)` puts the point test over an extension too large for the build's budget (the next step: certificates searched offline and only verified in the build), and 13 prime-power levels at ℓ `= 2, 3` with only an upper bound. The other two are on binary curves: ℓ `= 31` at depth 1 over `GF(2^47)` and ℓ `= 31543` over `GF(2^61)`.

The 14 curves whose Δ did not factor carry cofactors of 112 to 152 digits that ECM (to 35 digits) and the NFS budget here did not split.

Scope

Every decision is a proof from exact arithmetic (point certificates, closed or open walks with simple roots, class-group orders, Cornacchia, Deuring for class number 1); none is statistical. Unfactored cofactors of 112 to 152 digits are beyond the NFS budget of this container.